MT8002 Elements of Set Theory
The Zermelo–Fraenkel Axioms
Axioms 7 to 9

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Preliminaries

Textbook

Derek Goldrei (1996). Classic Set Theory. A Guided Independent Study.

Conventions

The numbers and page numbers assigned to chapters, examples, exercises, figures, quotes, sections and theorems on these slides correspond to the numbers assigned in the textbook.

See the appendix in the course homepage for the logical symbols, mathematical symbols and acronyms not defined in these slides.

Outline

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Introduction

Formal language

Recall that in ZFC the concepts of pure set and membership relation are primitive (undefined). Also recall that the universe of discourse of the theory is the pure sets.

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Classification

- Axioms stating the existence of sets
- Axioms determining properties of sets
- Axioms for building sets from other sets

Outline

The Axiom of Infinite

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$\exists x\,(\varnothing \in x \wedge \forall y\,(y \in x \rightarrow y \cup \{y\} \in x))$.

The Axiom of Infinite

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$\exists x\,(\emptyset \in x \wedge \forall y\,(y \in x \rightarrow y \cup \{y\} \in x))$.

Informal description

"*There is an inductive set.*" *(p. 92)*

Category

Axiom stating the existence of set.

Outline

The Construction of the Set of the Natural Numbers

Defining the natural numbers

For defining the natural numbers in ZFC we need to define zero and successor following the Dedekind–Peano axioms.

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Definition

A **numeral** $\mathbf{n}$ is the representation of a natural number n in a formal system.

The Construction of the Set of the Natural Numbers

Definition (p. 38)

$\mathbf{0} := \varnothing$.

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$$x^+ := x \cup \{ x \}.$$

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Exercise 4.21

Let x be a set. Show that z^+ is also a set.

The Construction of the Set of the Natural Numbers

Example

Some numerals.

$$
\begin{aligned}
\mathbf{0} & && && := \varnothing, \\
\mathbf{1} &:= \mathbf{0}^{+} \doteq \varnothing^{+} && \doteq \varnothing \cup \{\varnothing\} && \doteq \{\varnothing\}, \\
\mathbf{2} &:= \mathbf{1}^{+} \doteq \varnothing^{++} \doteq (\varnothing^{+})^{+} && \doteq \{\varnothing\} \cup \{\{\varnothing\}\} && \doteq \{\varnothing, \{\varnothing\}\}, \\
\mathbf{3} &:= \mathbf{2}^{+} \doteq \varnothing^{++} \doteq (\varnothing^{++})^{+} && \doteq \{\varnothing, \{\varnothing\}\} \cup \{\{\varnothing, \{\varnothing\}\}\} && \doteq \{\varnothing, \{\varnothing\}, \{\varnothing, \{\varnothing\}\}\}.
\end{aligned}
$$

The Construction of the Set of the Natural Numbers

Definition (p. 39)

A set y is **inductive** if

(i) $\varnothing \in y$ and

(ii) if $x \in y$ then $x^+ \in y$.

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(i) $\varnothing \in y$ and

(ii) if $x \in y$ then $x^+ \in y$.

Or, equivalently,

$$\frac{}{\varnothing \in y}, \qquad \frac{x \in y}{x^+ \in y}.$$

The Construction of the Set of the Natural Numbers

Definition (p. 40)

The **set of natural numbers**, denoted $\mathbb{N}$, is the intersection of all inductive subsets of any inductive set y, that is,

$$\mathbb{N} := \bigcap \{ z : z \text{ is an inductive subset of } y \},$$

Wrong! (Naive Comprehension Principle),

Right! (Axiom of Infinite),

$$\mathbb{N} := \bigcap \{ z \in \mathscr{P}(y) : z \text{ is inductive} \},$$

Right! (Axiom of Separation).

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Definition (p. 40)

A **natural number** is a member of $\mathbb{N}$.

Outline

The Axiom of Replacement

Axiom of Replacement

$\forall x \exists y \forall y' \, (y' \in y \leftrightarrow \exists x' \, (x' \in x \wedge \phi(x', y')))$,

where $\phi(s, t)$ is a formula (possibly referring to named sets (possibly with parameters)) such that $\forall s \exists t \, (\phi(s, t) \wedge \forall t' \, (\phi(s, t') \rightarrow t' \doteq t))$.

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Informal description

"*If $\phi(s, t)$ is a class function, then when its domain is restricted to [a set] x, the resulting images form a set y.*" *(p. 92)*

Category

Axiom for building a set from other set.

The Axiom of Replacement

Example

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$$\mathscr{A} := \{\, \langle s, s\rangle : s \text{ is a set and } \phi(s,s) \text{ holds} \,\}.$$

The Axiom of Replacement

Example

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For example, let $\phi(s,t)\colon\ s \doteq t$ be a formula (class function). From $\phi(s,t)$ we can define the proper class

$$\mathscr{A} := \{\, \langle s,s\rangle : s \text{ is a set and } \phi(s,s) \text{ holds} \,\}.$$

Note that $\mathscr{A}$ is a proper class because the universe of sets $\mathscr{V}$ is the proper class of first coordinates of the ordered pairs it contains.

The Axiom of Replacement

Exercise 4.23

Let x be a set. Show that $\{\{y\} : y \in x\}$ is a set.

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Solution

Let $\phi(s,t) \colon t \doteq \{s\}$ be a formula (class function).

By the Axiom of Replacement when the formula (class function) $\phi(s,t)$ is restricted to the set x there is a set formed with the images of the class function, that is, $\{\{y\} : y \in x\}$ is a set.

The Axiom of Replacement

Remark

Without the Axiom of Replacement we could not prove the existence of the set

$$\{\mathbb{N}, \mathscr{P}(\mathbb{N}), \mathscr{P}(\mathscr{P}(\mathbb{N})), \mathscr{P}(\mathscr{P}(\mathscr{P}(\mathbb{N}))), \ldots\}.$$

Outline

Axiom of Foundation

$\forall x\,(x \not\doteq \emptyset \;\rightarrow\; \exists y\,(y \in x \;\wedge\; x \cap y \doteq \emptyset))$.

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$\forall x\,(x \not\doteq \emptyset \;\rightarrow\; \exists y\,(y \in x \;\wedge\; x \cap y \doteq \emptyset))$.

Category

Axiom determining a property of sets.

The Axiom of Foundation

Exercise 4.41(a)

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Solution

The singleton $\{z\}$ is a set by the Axiom of Pairs whose unique element is z. By the Axiom of Foundation

$$\{z\} \cap z \doteq \varnothing.$$

Therefore, $z \notin z$ (otherwise $\{z\} \cap z \not\doteq \varnothing$).

The Axiom of Foundation

Definition

Let $\langle x_n \rangle_{n \in \mathbb{N}}$ be a sequence. An **infinite descending $\in$-chain** is an sequence such that $x_{n+1} \in x_n$, for all $n \in \mathbb{N}$, that is,

$$\cdots x_{n+1} \in x_n \in \cdots \in x_2 \in x_1 \in x_0.$$

The Axiom of Foundation

Theorem 4.3 (No infinite descending $\in$-chains)

Let $\langle x_n \rangle_{n \in \mathbb{N}}$ be a sequence. The sequence cannot form an infinite descending $\in$-chain.

The Axiom of Foundation

Theorem 4.3 (No infinite descending $\in$-chains)

Let $\langle x_n \rangle_{n \in \mathbb{N}}$ be a sequence. The sequence cannot form an infinite descending $\in$-chain.

Proof (by contradiction)

Suppose that $\langle x_n \rangle_{n \in \mathbb{N}}$ forms an infinite descending $\in$-chain.

Let $x = \{ x_n : n \in \mathbb{N} \}$. By the Axiom of Foundation exists y such that $x \cap y \doteq \emptyset$.

Since $y \in x$, let $y = x_n$ for some n. Since $x_{n+1} \in x$ and $x_{n+1} \in y$ then $x \cap y \not\doteq \emptyset$. Contradiction.

Outline

References

Derek Goldrei (1996). Classic Set Theory. A Guided Independent Study. Chapman & Hall (cit. on p. 2).